

40 marks • For teacher use after the assessment

Marking guidance

Award each distinct point once. Accept equivalent correct pseudocode and explanations. Credit follow-through where a single earlier error is carried consistently into a later trace, unless that later mark assesses the same error. No GCSE grade boundaries are assigned to this topic test.

1 Computational thinking [6]

- 1a Keep relevant details (1) and remove unnecessary detail for a purpose (1).
- 1b Suitable relevant detail, e.g. borrower ID / item ID / due date (1). Suitable irrelevant detail, e.g. favourite film (1).
- 1c Split into smaller sub-problems (1), making the parts easier to develop, understand or test (1).

2 Structure diagram [4]

Whole system above the three named main parts (1). Correct parent-child links (1). Check availability beneath Loans (1). Record loan beneath Loans (1). Accept equivalent appropriate loan sub-problems.

3 Searching [6]

3a Checks 2,4,6,8,10,12 (1), six comparisons (1).

3b First midpoint index 3 has value 8 (1). Keep indices 4 to 6 (1). Next midpoint index 5 has value 12, so found (1).

3c Data must be sorted by the search key (1).

4 Sorting [10]

4a [2,6,5,1] (1), [2,5,6,1] (1), [2,5,1,6] (1). Value 6 is in its final position (1).

4b Compare 2 and 3, take 2 (1). Compare 8 and 3, take 3 (1). Final [2,3,7,8] (1).

4c [2,5,4,1] (1), [2,4,5,1] (1), [1,2,4,5] (1).

5 Create a flowchart [5]

Input age in an input/output symbol (1). Decision $\text{age} \geq 12$ in a diamond (1). True and false branches correctly labelled (1). Correct output on each branch in input/output symbols (1). Connected start and end terminals with sensible flow lines (1).

6 Debug an algorithm [3]

6a Logic error (1).

6b $\text{average} = (a + b) / 2$ (1). Parentheses ensure addition happens before division (1).

7 Complete a trace table [6]

Initial total 0 (1). Totals after iterations: 1 (1), 3 (1), 6 (1), 10 (1). Final output 10 (1). Accept equivalent trace conventions that clearly distinguish variable state from output.